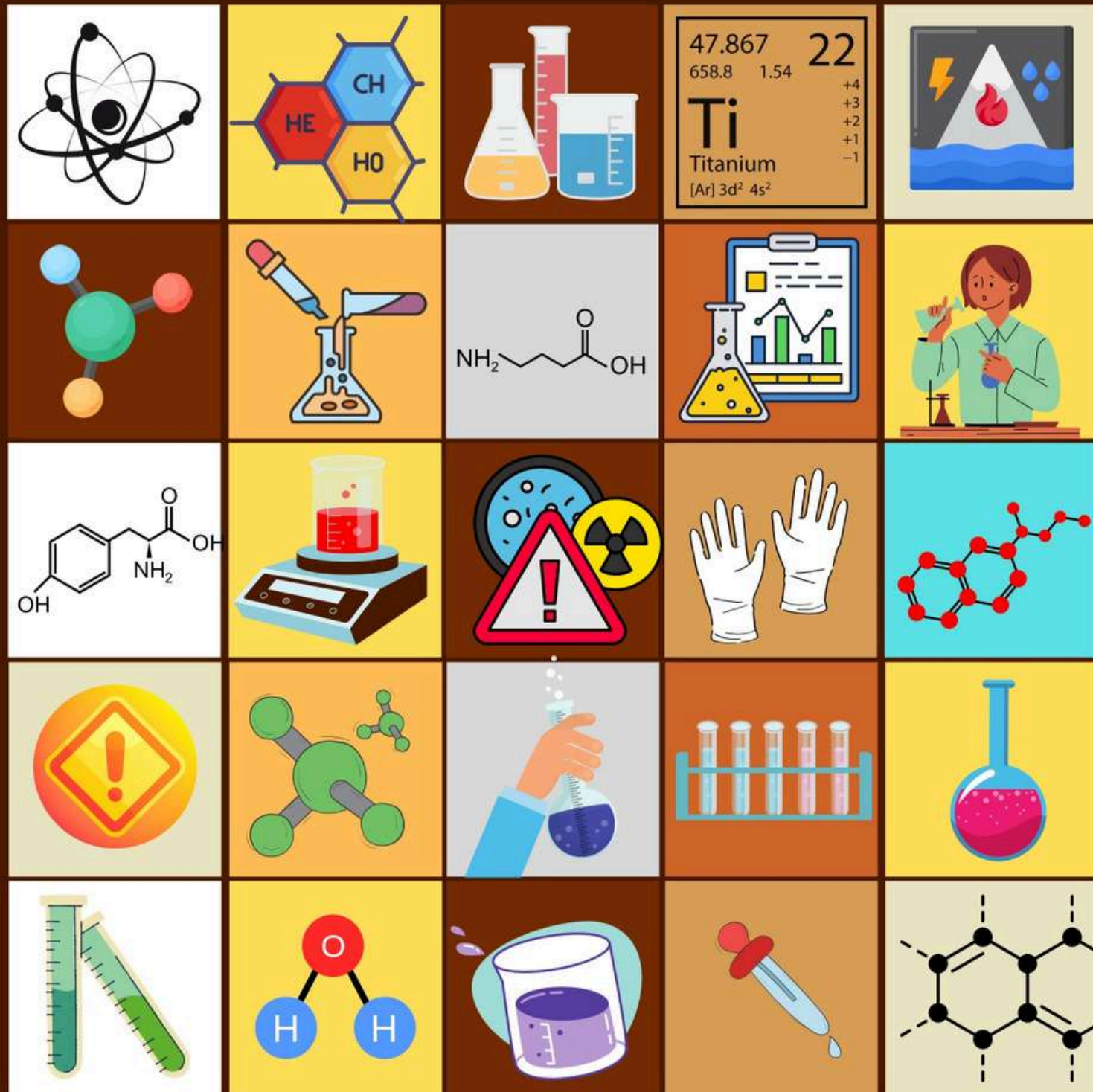




# CLASS 9th NOTES CHEMSITRY

## IS MATTER AROUND US PURE?

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# IS MATTER AROUND US PURE?



## Substance :

A substance is a kind of matter that cannot be separated into other kind of matter by any physical process.

## Pure Substance :

A pure substance is a form of matter that has a definite composition and distinct properties. They are made of only one entity.

e.g. Milk, water, diamond, etc.



## Mixture :

Mixtures are combinations of different substances, either elements or compounds, which retain their individual properties.

e.g. Crude Oil, Seawater, etc.

## Types of mixtures :-

**Homogeneous mixture :** A mixture which has a uniform composition throughout is called a homogeneous mixture or solution.

Examples: sugar in water, salt in water.

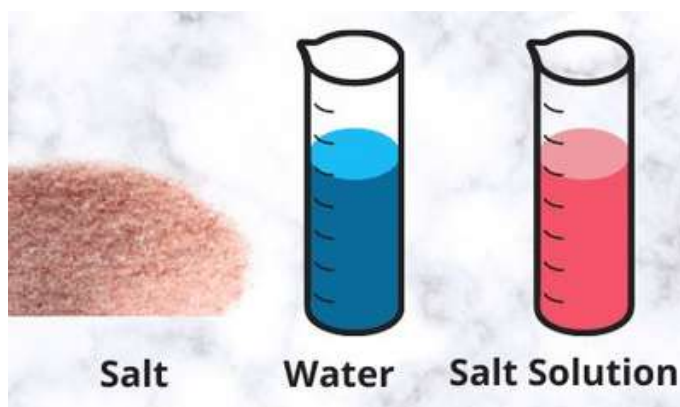
**Heterogeneous mixture :** A mixture which contains physically distinct parts and has a non-uniform composition is called a heterogeneous mixture.

Examples: Mixture of salt and iron filings, sand and sugar.

**Solution :** A solution is a homogeneous mixture of two or more substances. e.g. Soda water, Lemonade, etc.

### Components of a Solution:

- **Solute:** The substance that is dissolved in the solvent. It is present in a lesser amount.
- **Solvent:** The substance in which the solute is dissolved. It is present in a greater amount.



**Alloys** are mixtures of metals or a metal and a non-metal, unseparable by physical methods. Though not separable, they retain properties of constituents and can vary in compositions that's why they are considered as a mixture. Example: brass is 30% zinc and 70% copper.

### Properties of a solution:

1. A solution is a **homogeneous** mixture.
2. The particles of a solution are **smaller** than **1 nm** (10<sup>-9</sup> metre) in diameter. So, they cannot be seen by naked eyes.
3. Because of very small particle size, they **do not scatter a beam of light** passing through the solution. So, the path of light is not visible in a solution.
4. The **solute particles cannot be separated** from the mixture by the process of filtration. The solute particles do not settle down when left undisturbed, that is, a solution is stable

### Concentration of Solution :

**Mass by mass percentage**

$$= \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

**Mass by volume percentage**

$$= \frac{\text{Mass of solute}}{\text{Volume of solution}} \times 100$$

**Volume by volume percentage of a solution**

$$= \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

|   | SOLIDS                                                                     | LIQUIDS                                                                                                            | GASES                                                                             |
|---|----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1 | Size of solute particles is smallest. $< 10^{-9}\text{m}$                  | Size of solute particles bigger than true but smaller than suspension. In between $10^{-9}$ to $10^{-6}\text{m}$ . | Size of particles biggest. $> 10^{-6}\text{m}$                                    |
| 2 | Solute particles can't be seen with naked eye.                             | Solute particles can't be seen with naked eye.                                                                     | Can be seen with naked eye.                                                       |
| 3 | Homogeneous mixture.                                                       | Seems homogeneous but actually heterogeneous.                                                                      | Heterogeneous mixture.                                                            |
| 4 | Particles can't be separated by filtration .                               | Particles can't be separated by filtration .                                                                       | Can be separated by filtration.                                                   |
| 5 | Transparent                                                                | Translucent                                                                                                        | Opaque                                                                            |
| 6 | Stable solutions - i.e. solute particles do not settle on keeping.         | Stable solutions                                                                                                   | Unstable solution-solute particles settle upon keeping.                           |
| 7 | Do not show tyndall effect.                                                | Show tyndall effect.                                                                                               | May or may not show tyndall effect.                                               |
| 8 | Solutions diffuse rapidly through filter paper as well as parchment paper. | Colloid particles pass through filter paper but not through parchment paper.                                       | Suspension particles do not pass through filter paper as well as parchment paper. |
| 9 | e.g. Sugar in water.                                                       | e.g. Milk, blood.                                                                                                  | e.g. Sand/mud in water.                                                           |

**Suspension:** A suspension is a heterogeneous mixture in which the solute particles do not dissolve but remain suspended throughout the bulk of the medium basically suspension is non - homogeneous systems in which solids are dispersed in liquids.

#### Properties of a suspension:

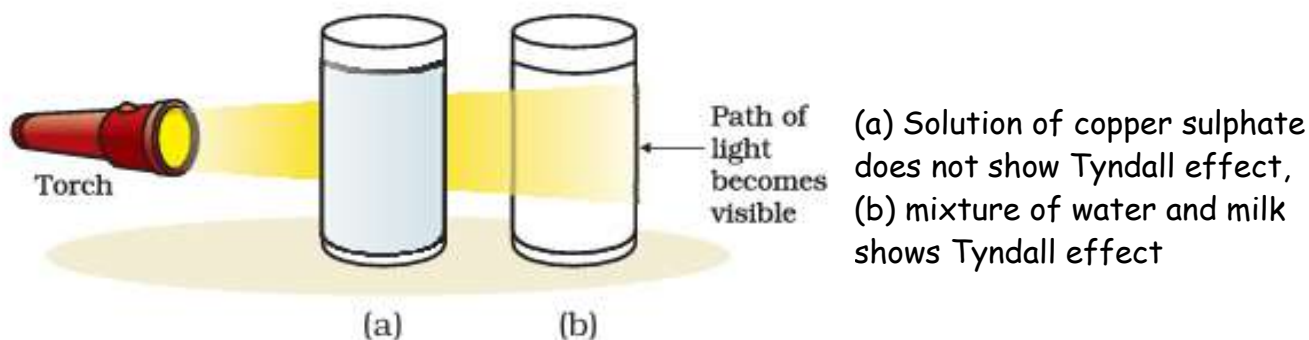
- Suspension is a **heterogeneous** mixture.
- Particles are **visible** to the naked eye.

- Particles scatter light, making the **path visible**.
- Solute particles settle over time, making the **suspension unstable**.
- Particles can be separated by **filtration**.
- When particles settle, the suspension breaks and no longer scatters light.

### Colloidal Solution

The mixture in which one substance of dispersed insoluble particles is suspended throughout the other substance is known as colloidal solution or colloids.

The phenomenon of scattering of a beam of light by colloidal particles, when passed through a colloidal solution, is called the **Tyndall effect**.



### Properties of Colloidal solution:

1. A colloid is a **heterogeneous** mixture.
2. Particle size is **too small** to be seen by naked eyes.
3. Particles scatter light, making its **path visible**.
4. Colloids are stable and **do not settle** when left undisturbed.
5. Cannot be separated by filtration; requires centrifugation.

### Common examples of colloids :

**Dispersed Phase** : The solute-like component of the dispersed particles in a colloid form the dispersed phase.

**Dispersion Medium** : The component in which the dispersed phase is suspended is known as the dispersing medium.

**Aerosol** : A colloidal solution with dispersed phase solid/liquid and dispersing medium gas is called Aerosol. e.g. clouds.

**Foam** : A colloidal solution with dispersed phase gas and dispersing medium solid/liquid is called Foam. e.g. Shaving cream.

| Dispersed phase | Dispersing Medium | Type      | Example                        |
|-----------------|-------------------|-----------|--------------------------------|
| Liquid          | Gas               | Aerosol   | Fog, clouds, mist              |
| Solid           | Gas               | Aerosol   | Smoke, automobile exhaust      |
| Gas             | Liquid            | Foam      | Shaving cream                  |
| Liquid          | Liquid            | Emulsion  | Milk, face cream               |
| Solid           | Liquid            | Sol       | Milk of magnesia, mud          |
| Gas             | Solid             | Foam      | Foam, rubber, sponge, pumice   |
| Liquid          | Solid             | Gel       | Jelly, cheese, butter          |
| Solid           | Solid             | Solid Sol | Coloured gemstone, milky glass |

Ye table bhi dekhlena



## Physical and Chemical Changes:

| Physical change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Chemical change                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. In a physical change, only physical properties such as colour, physical state, density, volume, etc. change; chemical properties remain unchanged.</li> <li>2. No new substance is formed in a physical change.</li> <li>3. Very little or no energy in the form of heat, light or sound is usually absorbed or given out in a physical change.</li> <li>4. A physical change is a temporary change.</li> <li>5. The original form of substance can be regained by simple physical methods.</li> <li>6. A physical change is reversible.</li> </ol> | <ol style="list-style-type: none"> <li>1. In a chemical change, the chemical composition and chemical properties undergo a change.</li> <li>2. A new substance is formed in a chemical change.</li> <li>3. A chemical change is always accompanied by absorption or evolution of energy.</li> <li>4. A chemical change is a permanent change.</li> <li>5. Original substance cannot be obtained by simple physical methods.</li> <li>6. A chemical change is irreversible.</li> </ol> |

e.g. ice, water, and water vapor

e.g. Oil burns in air, water extinguishes fire.

## Types of Pure Substances:

**1 Elements** - Basic form of matter that cannot be broken down by chemical reactions.

- **Categories:** Metals, non-metals, metalloids.

| Property     | Metals                                  | Non-Metals                                                     |
|--------------|-----------------------------------------|----------------------------------------------------------------|
| Lustre       | Shiny (e.g., gold, silver)              | Not shiny (e.g., carbon, sulfur)                               |
| Color        | Silvery-grey or golden-yellow           | Various colors (e.g., oxygen - colorless, iodine - purple)     |
| Conductivity | Good conductors of heat and electricity | Poor conductors of heat and electricity (e.g., oxygen, sulfur) |
| Ductility    | Ductile (e.g., copper, gold)            | Not ductile                                                    |
| Malleability | Malleable (e.g., gold, aluminum)        | Not malleable (e.g., sulfur)                                   |
| Sonorous     | Sonorous (e.g., copper, brass)          | Non-sonorous                                                   |



" Burning a candle involves both physical (melting wax) and chemical changes (wax burning)."

Some elements have intermediate properties between those of metals and nonmetals, they are called **metalloids**; examples are boron, silicon, germanium, etc

**Compounds** : A compound is a substance composed of two or more elements, chemically combined with one another in a fixed proportion.

| Mixtures                                                                                    | Compounds                                                                           |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1. Elements or compounds just mix together to form a mixture and no new compound is formed. | 1. Elements react to form new compounds.                                            |
| 2. A mixture has a variable composition.                                                    | 2. The composition of each new substance is always fixed.                           |
| 3. A mixture shows the properties of the constituent substances.                            | 3. The new substance has totally different properties.                              |
| 4. The constituents can be separated fairly easily by physical methods.                     | 4. The constituents can be separated only by chemical or electrochemical reactions. |